

§1

$$\Delta = AC - B^2 > 0 \quad A > 0 \text{ 极大}$$

$$\Delta < 0 \text{ 极大}$$

$$\Delta = 0 \text{ 不确定}$$

$$\Delta < 0 \text{ 鞍点}$$

$$d\rho = |\mathbf{r}|^2 dr$$

$$= |\mathbf{r}|^2 \cdot 4\pi r dr$$

$$= \left(\frac{1}{\sqrt{\pi} a_0} e^{-r/a_0} \right)^2 \cdot 4\pi r dr$$

$$= \frac{1}{\pi a_0^3} e^{-2r/a_0} \cdot 4\pi r dr$$

$$= \frac{4r}{a_0^3} e^{-2r/a_0} dr$$

symmetric energy

$$k = \sqrt{\frac{2mE}{\hbar^2}}$$

$$ka = \frac{n}{2}\pi$$

$$k = \frac{n\pi}{2a}$$

$$E = \frac{\hbar^2 k^2}{2m}$$

$$\frac{d}{dx} \int_0^x (x-t) f(t) dt = \int_0^x f(t) dt - \frac{d}{dx} \int_0^x t f(t) dt$$

$$= \int_0^x f(t) dt - x f(x)$$

Day 3-811

$$-\frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + \delta(x) \psi = E \psi$$

$$-\int_{-\epsilon}^{\epsilon} \frac{\hbar^2}{2m} \frac{d^2 \psi}{dx^2} + \int_{-\epsilon}^{\epsilon} \delta(x) \psi dx = \int_{-\epsilon}^{\epsilon} E \psi dx$$

$$\frac{2m\delta(x)}{\hbar^2} \psi(0) = \psi'(0)$$

$$\S 2 \quad MI \quad f(x, y) = x^3 - y^3 - 3x^2 + 3y^2$$

$$f_x = 3x^2 - 6x$$

$$f_y = -3y^2 + 6y$$

$$f_x = 0 \Rightarrow \begin{cases} x_1 = 2 \\ x_2 = 0 \end{cases}$$

$$\text{驻点: } (0, 0)$$

$$(0, 2)$$

$$(2, 0)$$

$$(2, 2)$$

$$f_y = 0 \Rightarrow \begin{cases} y_1 = 2 \\ y_2 = 0 \end{cases}$$

$$A = f_{xx} = 6x - 6$$

$$B = f_{xy} = 0$$

$$C = f_{yy} = -6y + 6$$

$$\Delta = AC - B^2 = -36xy + 36x + 36y - 36$$

$$= (x+y-1) \cdot 36$$

$$\textcircled{3} (0, 0) \quad A < 0 \quad \Delta < 0 \text{ 鞍点}$$

$$(0, 2) \quad A < 0 \quad \Delta > 0 \text{ 极大值}$$

$$(2, 0) \quad A > 0 \quad \Delta > 0 \text{ 极大值}$$

$$(2, 2) \quad A > 0 \quad \Delta < 0 \text{ 鞍点}$$

M2

$$u = xyz$$

$$\text{构 } L = xyz - \lambda \left(\frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1 \right)$$

$$\frac{\partial L}{\partial x} = yz - \frac{2\lambda x}{a^2}$$

$$\frac{\partial L}{\partial y} = xz - \frac{2\lambda y}{b^2}$$

$$\frac{\partial L}{\partial z} = xy - \frac{2\lambda z}{c^2}$$

$$\frac{\partial L}{\partial \lambda} = \frac{x^2}{a^2} + \frac{y^2}{b^2} + \frac{z^2}{c^2} - 1$$

·x

·y

·z

$$\Rightarrow \frac{x}{a^2} = \frac{y}{b^2} = \frac{z}{c^2} = \frac{1}{3}$$

$$\Rightarrow u_{\max} = \frac{1}{27} abc$$

Q1

1. $k \equiv \sqrt{\frac{2mE}{\hbar^2}}$

$ka = \frac{n\pi}{2}$

$k = \frac{n\pi}{2a}$

$E = \frac{\hbar^2 k^2}{2ma^2}$

$\psi(x) = \psi(x)$ 4 有确定的宇称. $n=1$ 偶宇称. $\psi_1 = A \cos(\frac{\pi}{2a}x)$.

$n=2$ 奇宇称. $\psi_2 = A \sin(\frac{\pi}{a}x)$.

$\int_{-a}^a \psi^2 dx = \int_{-a}^a A^2 \cos^2(\frac{\pi}{2a}x) dx = A^2 \int_{-a}^a (1 + \cos \frac{\pi x}{a}) dx = A^2 \cdot a + \frac{1}{\pi} \sin \frac{\pi x}{a} \Big|_{-a}^a = A^2 \cdot a$
 $\Rightarrow A = \sqrt{\frac{1}{a}}$

$\psi_1 = \sqrt{\frac{1}{a}} \cos(\frac{\pi}{2a}x)$

$\psi_2 = \sqrt{\frac{1}{a}} \sin(\frac{\pi}{a}x)$

$\langle p \rangle = \langle \psi | \hat{p} | \psi \rangle = \int \psi^* (-i\hbar \frac{\partial}{\partial x}) \psi dx = \frac{-i\hbar}{2a} \int_{-a}^a \cos(\frac{\pi}{2a}x) \sin(\frac{\pi}{2a}x) dx = 0$
 $\frac{i\hbar}{2a} \int_{-a}^a \sin \frac{\pi x}{2a} \cos \frac{\pi x}{2a} dx = 0$

II

$k = ka = \sqrt{\frac{2mE}{\hbar^2}} \cdot a$

$\eta = K a = \sqrt{\frac{2m(V-E)}{\hbar^2}} \cdot a$

$k^2 + \eta^2 = \frac{2mV a^2}{\hbar^2} = R^2$
 $\eta = R \tan \delta$

III

3 个束缚态.

$\frac{\pi}{2} < R \leq \frac{3\pi}{2}$
 $\frac{\hbar^2 k^2}{2ma^2} < V_0 < \frac{9\hbar^2 k^2}{8ma^2}$

Q2

I. $\mu = \frac{mM}{m+M}$ $l \rightarrow 0$ $r \rightarrow 0$ $\sigma \propto r^2 > \frac{\hbar^2}{2}$ $p \rightarrow \infty$ $E \rightarrow \infty$

$V_{eff} = -\frac{e^2}{r} - \frac{\hbar^2}{2m} \cdot \frac{1}{r^2} = -\frac{e^2}{r} + \frac{\hbar^2 l(l+1)}{2mr^2}$

II. $R_{nl} = \frac{1}{r} \psi_{nl}$ $\psi_{nl} = \frac{4r^2}{a_0^3} e^{-2r/a_0}$

$\frac{dr}{dt} = \frac{dr}{dt} e^{-2r/a_0} + \frac{4r^2}{a_0^3} \cdot \frac{-2}{a_0} e^{-2r/a_0} = 0$
 $r = a_0$

III $w = \int_0^\infty p(r) dr = \int_0^\infty r^2 e^{-2r/a_0} dr$